Predicting Molecular Shape from Electron Pairs

PhET Molecule Shapes · PhET

Senior Secondary (Years 11-12)	~30 minutes	Chemistry

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Simulation: https://phet.colorado.edu/en/simulations/molecule-shapes
Curriculum links: bonding · chemistry · organic chemistry

Learning intentions

- I can predict a molecule's 3D shape from its number of bonding and lone electron pairs.
- I can explain how lone pairs affect bond angles compared with bonding pairs alone.
- I can relate VSEPR theory to the structure of real molecules.

Getting started

Open Molecule Shapes and start on the Model screen. Before revealing the shape name for each set-up, predict and sketch or describe what you expect based on the number of bonding and lone pairs you've added to the central atom.

Tasks

1. Add two bonding pairs and no lone pairs to the central atom. Predict the shape and bond angle before revealing it, then record the actual result.

2. Build molecules with 2, 3 and 4 bonding pairs (no lone pairs) and record the resulting shape and bond angle for each.

Bonding pairs	Shape	Bond angle

3. Add one lone pair to a molecule with three bonding pairs. What happens to the shape's name and the bond angle compared with having no lone pairs?

4. Explain, using electron-pair repulsion, why lone pairs push bonding pairs closer together and reduce the bond angle.

5. Build a molecule with two bonding pairs and two lone pairs. Predict its shape before revealing the result, then compare your prediction with what the simulation shows.

6. The ideal tetrahedral bond angle (four bonding pairs, no lone pairs) is 109.5°. Using what you found about lone pairs reducing bond angles, estimate whether the H-N-H angle in ammonia (one lone pair) would be smaller or larger than the H-C-H angle in methane (no lone pairs), and explain your reasoning.

Working:

7. Choose one real molecule — water, ammonia or methane — and explain its actual shape in terms of its electron pairs, using what you found in the simulation.

Extension (optional)

Explain how the polarity of a molecule (whether its individual bond dipoles cancel out or not) depends on both its bonding pattern and its overall 3D shape, using one example from this simulation.